

## 6. $I^2$ corrects Q by removing the scatter chance is expected to produce

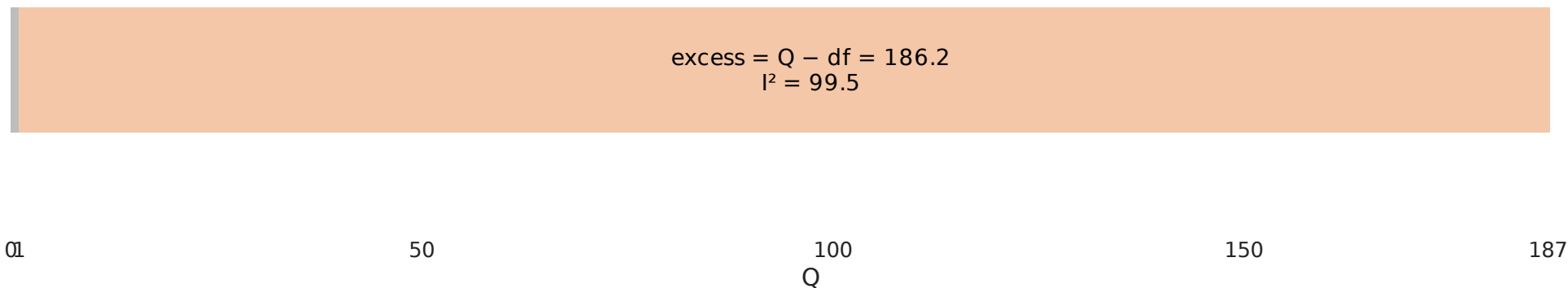
$$I^2 = \max(0, (Q - df) / Q) \times 100.$$

The numerator drops df, the average Q when sampling error is the only difference.

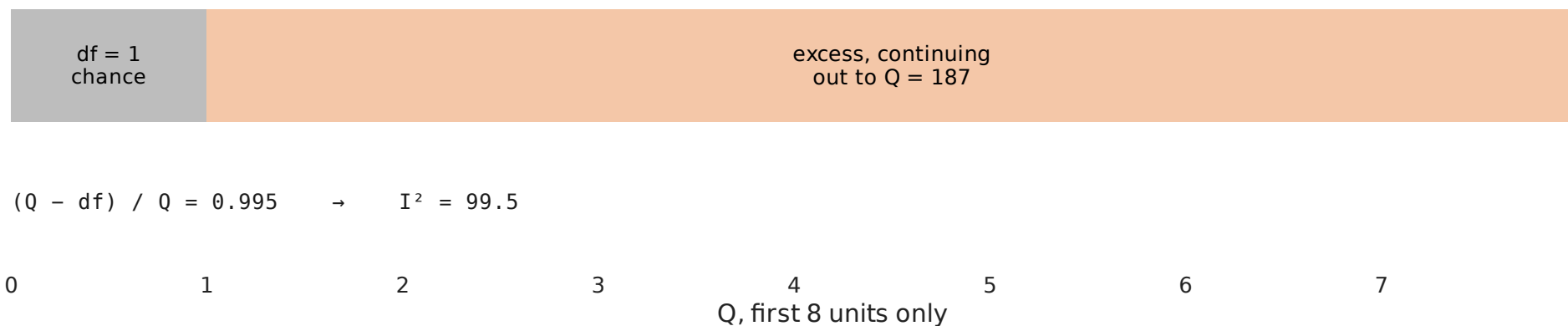
What remains is a percent of Q. Here that percent is 99.5.

### PGS000004, the real Q

**Whole bar = observed Q = 187.2. Grey edge at the left = df = 1.**



### Left edge of the same bar, zoomed



$I^2$  does not move the pooled AUC. The pooled AUC stays 0.645.

$I^2 = 99.5$  means almost all of Q is heterogeneity beyond sampling error, so this European cell is removed by the  $I^2$  filter before Figure 1C.

The eMERGE logit, 0.405, is the European number in the Stage-2 scale check. Q here is the within-ancestry scatter, computed after the second study is included.